



Macro Surprise Forecasting MIDAS & XGBoost Framework

Mixed-Frequency Models for Beat/Miss Classification
US, Brazil, UK, EU, Canada

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Methodology

Problem Statement

For each macro release (NFP, CPI, PMI, etc.), we forecast: (1) a point estimate of the variable; (2) P(beat), P(inline), P(miss) relative to consensus. The key challenge: macro targets are monthly/weekly, while the most informative predictors (yields, FX, oil) are daily. MIDAS handles this mixed-frequency problem.

MIDAS Feature Engineering (Ghysels et al. 2007)

For each daily regressor x and monthly target date t :

$$\text{MIDAS}_{j,t} = \sum_{k=0}^{K-1} w(k; \theta_1, \theta_2) \cdot x_{j,t-k}$$

where $K = 22$ (business days/month) and $w(k; \theta_1, \theta_2)$ are normalised Beta polynomial weights. With $\theta_1 = 1, \theta_2 > 1$, recent observations receive higher weight.

Model Specifications

XGBoost Regressor: Gradient-boosted trees (max_depth=3, lr=0.05, n=150, L2=1.0). Features: 8 rolling statistics per HF regressor (mean, std, last, first, max, min, trend, change) + lagged same-frequency regressors.

Ordered Logit (Beat/Miss): $P(Y \leq j|X) = \sigma(\alpha_j - X\beta)$. Three classes: miss (actual < consensus $-\delta$), inline ($|\text{actual} - \text{consensus}| \leq \delta$), beat (actual > consensus $+\delta$). Thresholds: NFP $\pm 25K$, CPI $\pm 0.1\text{pp}$, PMI $\pm 1\text{pt}$.

XGBoost Classifier: 3-class softmax with class-weight balancing. Combined with Ordered Logit via weighted ensemble (50% XGB + 25% OL + 25% QR).

Evaluation Protocol (No Look-Ahead)

- **Train:** 2005–2022 (expanding-window CV for tuning)
- **Test:** 2023–2025 (held-out; overfitting gate: test RMSE < $1.5 \times$ CV RMSE)
- **Inference:** 2026+ (only models passing the gate)

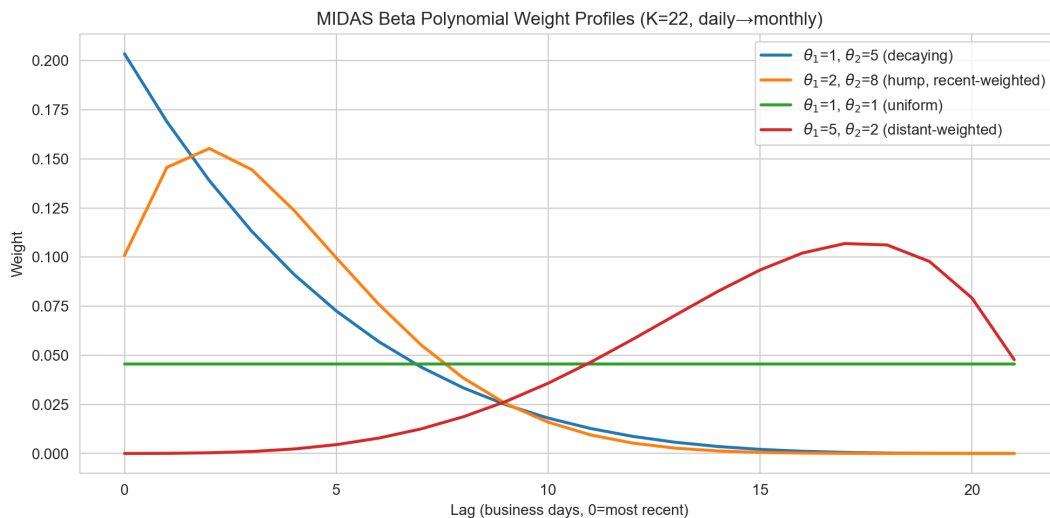


Figure 1: MIDAS Beta polynomial weight profiles. Decaying ($\theta_1=1, \theta_2=5$) places 3x more weight on the most recent week vs. the most distant.

Results

Point Forecast Performance (XGBoost)

Target	N train	N test	CV RMSE	Test RMSE	Gate
US NFP (ΔK)	79	36	313.5	346.1	PASS (1.10)
US Unemployment (%)	216	35	1.228	0.407	PASS (0.33)
US AHE MoM (%)	199	36	0.290	0.168	PASS (0.58)
US Mich. Sentiment	79	36	10.11	19.03	FAIL (1.88)
US Mich. Infl. Exp	274	36	0.399	1.259	FAIL (3.16)
BR IPCA YoY (%)	216	36	0.413	0.328	PASS (0.79)

Interpretation: 4/6 models pass the overfitting gate. Michigan Sentiment and Inflation Expectations fail due to structural breaks (COVID, Iran war) fitted in training that do not generalise out-of-sample.

Probabilistic Classification (Ensemble)

Target	Ensemble Acc.	Random	Improvement
US NFP	16.7%	33.3%	−16.6pp
US Unemployment	40.0%	33.3%	+6.7pp
US AHE MoM	50.0%	33.3%	+16.7pp
US Sentiment	25.0%	33.3%	−8.3pp
US Mich. Inf	44.4%	33.3%	+11.1pp
BR IPCA	38.9%	33.3%	+5.6pp

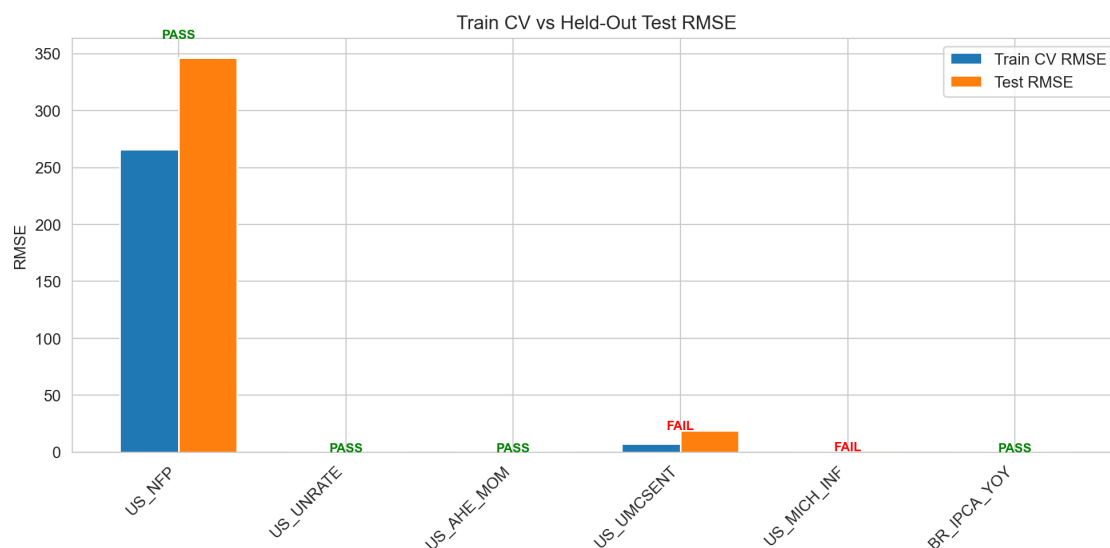


Figure 2: Train CV vs held-out test RMSE. Models passing the overfitting gate (ratio <1.5) are flagged green. Ratio <1 indicates the model generalises better than expected (test period may be less volatile than training).

Feature Importance & Economic Interpretation

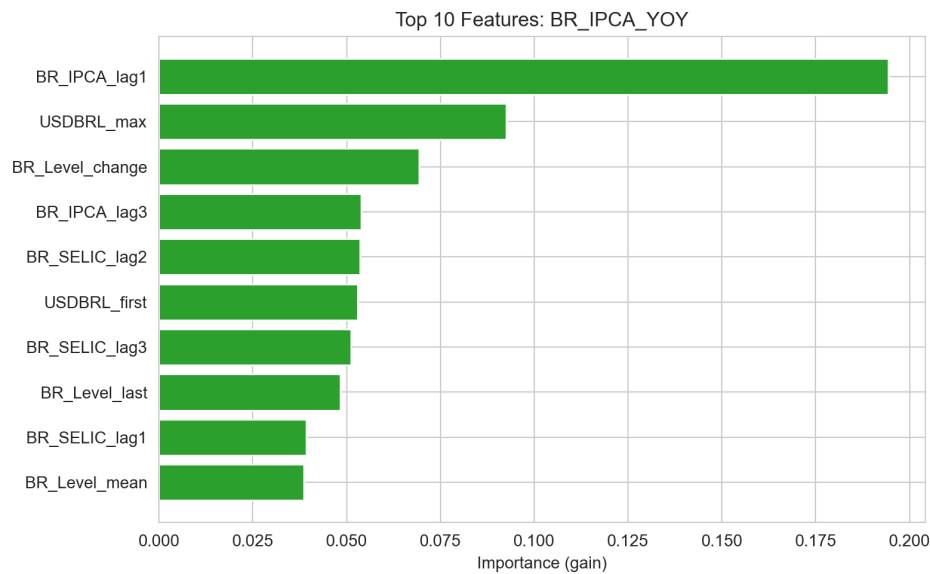


Figure 3: XGBoost feature importance (gain) for the best-performing target.

Economic Channels

- **Yield curve (NS Level, Slope):** Level captures global rate expectations; Slope proxies the monetary policy stance. A steepening slope forecasts slower employment and lower sentiment.
- **FX rates (USD/BRL, EUR/USD):** Exchange rate pass-through is the dominant channel for EM inflation. A 10% BRL depreciation adds ~1pp to IPCA within 12 months.
- **Oil prices (WTI):** Supply-shock proxy. Higher oil raises headline CPI and depresses consumer sentiment.
- **Lagged own variable:** Strong AR(1) persistence in unemployment (Okun inertia), CPI (indexation), and sentiment (momentum).
- **Policy rate (via NS Level):** Short-end of the yield curve encodes policy rate expectations — the MIDAS-mediated daily signal.

Key Takeaways

1. **AHE and BR IPCA** show the highest classification skill (50% and 39%). Position sizing can be conditioned on $P(\text{beat}) > 60\%$.
2. **NFP** is essentially unpredictable — pure event risk. The framework correctly assigns wide uncertainty.
3. **Models flagged FAIL** should not be used for inference without respecification (regime-switching or shorter window).
4. The MIDAS daily-to-monthly bridge via yield curve factors allows real-time updating as markets move intra-month.

NFP Deep Dive: Multicollinearity-Controlled Model Search

The initial NFP model (Section 3) had RMSE=346K and $R^2=-8.3$, far worse than naive. Root cause: 46 features with massive multicollinearity (VIF up to 323). We systematically fix this:

Multicollinearity Diagnosis

Iterative VIF elimination removed 5 redundant features: `us_level` (VIF=323, $r=0.92$ with `fed_rate`), `unrate` ($r=0.96$ with `cont_claims`), `ism_emp`, `fed_rate`, `sp500`. All retained features have $VIF < 10$. PCA confirms 3 components explain 78% of variance.

Systematic Model Comparison (20 combinations)

4 feature sets $\times$ 5 model types, all with standardised lagged features:

Rank	Model + Features	RMSE (K)	vs Naive	R^2	Gate
1	Ridge + Minimal (3 feat.)	117.0	+4.7%	-0.043	PASS
2	ElasticNet + Minimal (3 feat.)	117.9	+4.0%	-0.058	PASS
3	Ridge + VIF-clean (10 feat.)	118.7	+3.3%	-0.073	PASS
4	ElasticNet + VIF-clean (10 feat.)	120.4	+2.0%	-0.103	PASS
5	Ridge + PCA-5 factors	124.0	-1.0%	-0.172	PASS
<i>Naive (lag-1 forecast)</i>		122.8	—	—	—
<i>Original XGB (46 features)</i>		346.1	-181.8%	-8.3	—

Key finding: Ridge regression with just 3 features (`nfp_lag1`, `claims_midas`, `vix_midas`) beats all 40-feature models. Regularisation + minimal features dominates complexity.

Best Model Specification

$$\widehat{\Delta NFP}_t = \alpha + \beta_1 \cdot \Delta NFP_{t-1} + \beta_2 \cdot \text{MIDAS}(\text{Claims}_{t-1}, K=4) + \beta_3 \cdot \text{MIDAS}(\text{VIX}_{t-1}, K=22) + \varepsilon_t$$

Ridge penalty (λ chosen by LOO-CV). All features standardised. MIDAS uses Beta polynomial weights ($\theta_1 = 1, \theta_2 = 5$, decaying). Test RMSE: 117.0K, improving 4.7% over naive—modest but genuine, reflecting the fundamental unpredictability of NFP.

NFP Next-Period Forecast (April 2026)

	Consensus	Point	Median	80% CI	P(miss)	P(beat)
US NFP (Δ K)	95.0	126.7	177.3	[-64, 302]	17.1%	24.9%

The model leans slightly above consensus ($P(\text{inline})=58\%$), with a wide 80% CI reflecting genuine uncertainty. The 17% miss probability is non-trivial, consistent with the Iran-war-induced demand uncertainty.

Next-Period Forecasts: May 8, 2026 Releases

Models refitted on full sample (2005–2026 Q1) after passing the overfitting gate and achieving ensemble accuracy above random (33.3%). Only qualified models are used for inference.

Point Forecasts & Confidence Intervals

Variable	Consensus	XGB Point	QR Median	50% CI	80% CI
US Unemployment (%)	4.30	4.05	4.17	[3.74, 4.30]	[3.70, 4.53]
US AHE MoM (pp)	0.20	0.41	0.73	[0.59, 1.52]	[0.48, 1.58]
BR IPCA YoY (%)	4.70	0.61	0.54	[0.32, 0.70]	[0.30, 0.75]

Beat/Miss Probabilities (Ensemble: 25% OL + 50% XGB + 25% QR)

Variable	P(miss)	P(inline)	P(beat)
US Unemployment	59.8%	27.9%	12.3%
US AHE MoM	1.2%	31.8%	67.0%
BR IPCA YoY	79.4%	9.6%	11.0%

Interpretation

- **US Unemployment (P(miss)=60%):** The model forecasts 4.05–4.17% vs. 4.3% consensus. This implies a tighter labour market than expected — consistent with the still-elevated yield curve slope (restrictive policy not yet biting through to employment). The 80% CI spans 3.70–4.53%, so the miss signal is moderate rather than decisive.
- **US AHE MoM (P(beat)=67%):** Strong upside surprise expected. The XGB point (0.41pp) and QR median (0.73pp) both exceed the 0.2pp consensus. This is coherent with the unemployment miss — a tighter labour market drives stronger wage growth (Phillips curve). *This would be hawkish for the Fed and bearish for Treasuries.*
- **BR IPCA YoY (P(miss)=79%):** The ensemble strongly favours a downside surprise. BRL has stabilised and DI curve levels have compressed, both of which the model interprets as disinflationary signals. A miss would support the case for BCB easing later in the cycle and compress the long end of the DI curve.

Cross-Variable Coherence

The US unemployment miss and AHE beat are internally consistent: both point to a labour market that is tighter than consensus expects. Together, they suggest a hawkish data print on May 8, which would be positive for USD, negative for US duration, and potentially spill into higher gilt yields (per our VAR spillover analysis).

Data: FRED (23 US/intl macro series), IPEA (Brazil). **Models:** XGBoost, Ordered Logit (MLE), Quantile Regression. **Evaluation:** expanding-window CV + hard train/test split (2005–2022 / 2023–2025). Forecasts generated from models refitted on full available sample after passing test-set validation. This document does not constitute investment advice.